



IMDB

Ratings and Reviews for Movies and TV Shows

TEAM 4

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Team 4 - IMDb: Ratings and Reviews for Movies and TV Shows

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1. Introduction

1.1 Purpose

The purpose of this project is to develop an application that allows users to browse, rate, and review movies and TV shows. Users can register an account, search for titles, and contribute their own ratings/reviews. Admins will have additional privileges to manage content and user accounts.

1.2 Functions

The scope of this system includes:

- Providing a catalog of movies and TV shows.
- Enabling users to add ratings and reviews.
- Managing user accounts and user-generated content.

1.3 Overview

The application aims to be user-friendly and scalable, ensuring that users can easily find information on their favorite movies/TV shows and share their opinions with others. The system will store and display average ratings and user reviews, helping visitors decide what to watch.

2. User Requirements

1. Users

- Shall be able to search for movies/TV shows by title, genre, or release year.
- Shall be able to view average ratings and read reviews.
- Shall be able to create an account using an email address and password.
- Shall be able to log in and log out securely.
- Shall be able to submit a numeric rating (e.g., 1–10) for a movie/TV show.
- Shall be able to write textual reviews for a movie/TV show.
- Shall be able to edit or delete their own reviews.

- Shall be able to maintain a personal watchlist of titles.

2. Admins

- Shall be able to add and delete movies and TV shows
 - Shall be able to update movies and TV show details (title, release year, cast, etc.).
 - Shall be able to delete any review that violates policies.
 - Shall be able to manage user accounts (suspension, deletion, etc.).
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3. Functional Requirements

1. Search for a Movie/TV Show

- **Description/Action:**

The system must enable any User to perform a keyword-based search for movies or TV shows using criteria such as title, genre, or release year. The system must validate inputs and handle empty or invalid queries gracefully.

- **Requirements/Inputs:**

- User-entered keyword(s) or filter parameters (title, genre, release year).
- Input validation to ensure non-empty, properly formatted entries.

- **Source:**

- Input via the search bar on the homepage or dedicated search page.
- The search bar includes filtering criteria

- **Pre-condition:**

- The user is on a page with an active search field.
- The movie/TV show database is populated with the necessary metadata.

- **Post-condition:**

- The system processes the query, retrieves matching results, or returns a “No results found” message if applicable.
 - **Output:**
 - A list of matching movies/TV shows with basic details (e.g., title, release year, poster image) or an error message/prompt if the input is invalid.
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2. View Movie/TV Show Details (Including Ratings & Reviews)

- **Description/Action:**

The system must display a detailed page for a selected movie/TV show. This page must include key information (cast, etc.), the computed average rating, and user reviews. This function is available to all users, but the data shown may differ based on the role (e.g., additional moderation details for admins).
- **Requirements/Inputs:**
 - Selection of a movie/TV show from search results or a list.
 - Retrieval of detailed metadata, ratings, and reviews from the database.
- **Source:**
 - The database records for the chosen title along with the associated user-submitted content.
- **Pre-condition:**
 - A valid title is selected that exists in the database.
 - At least one rating or review exists; otherwise, a “No reviews yet” message is displayed.
- **Post-condition:**
 - The system collates available reviews.
- **Output:**
 - A comprehensive details page displaying title information, average rating (e.g., “7.5/10”), and the list of user reviews.

- For admin users, additional administrative flags or moderation status may be displayed.
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3. Submit a Numeric Rating

- **Description/Action:**

The system must allow the User to submit a numeric rating for a movie/TV show. The rating must be validated (ensuring it falls within the accepted range of 1–10), and any invalid input must trigger an error message.

- **Requirements/Inputs:**

- User's numeric input via a dropdown or input field, ensuring the value is numeric and between 1 and 10.
- Immediate input validation for range and format.

- **Source:**

- The rating input provided on the movie/TV show detail page.

- **Pre-condition:**

- The selected movie/TV show exists in the database.

- **Post-condition:**

- If valid, the rating is stored in the database, and the average rating is recalculated; if invalid, the system rejects the submission and displays a descriptive error message.

- **Output:**

- A confirmation message such as "Your rating has been submitted", along with the updated average rating displayed on the details page.
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4. Write a Textual Review

- **Description/Action:**

The system must allow the User to submit a textual review for a specific movie/TV show.

Input must be validated for non-emptiness and adherence to content guidelines.

- **Requirements/Inputs:**
 - User-entered review text via a dedicated review submission form.
 - Validation to ensure the text meets minimum/maximum length and content policies.
 - **Source:**
 - The review submission form on the movie/TV show detail page.
 - **Pre-condition:**
 - The user has navigated to a valid movie/TV show detail page.
 - **Post-condition:**
 - If the review meets validation criteria, it is stored in the database; if not, the system returns an error prompting corrections.
 - **Output:**
 - The newly submitted review is displayed under the reviews section, along with a confirmation message (“Your review has been submitted”).
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5. Maintain a Personal Watchlist

- **Description/Action:**

The system must allow the User to add movies/TV shows to their personalized watchlist. The system must check that each title is valid and not already present in the watchlist, handling duplicate entries or errors appropriately.
- **Requirements/Inputs:**
 - User’s action (clicking “Add to Watchlist”) from the movie/TV show detail page.
 - Input validation to ensure the title isn’t already in the watchlist.
- **Source:**
 - The movie/TV show detail page and the user’s account profile data.
- **Pre-condition:**

- The selected movie/TV show exists and is not already in the user's watchlist.
 - **Post-condition:**
 - The title is added to the user's watchlist in the database; if already present, an informative error message is returned.
 - **Output:**
 - The updated watchlist displayed on the user's profile page along with a confirmation message ("Title successfully added to your watchlist") or an error message if the action fails.
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6. Moderate User-Submitted Content (Admin Function)

- **Description/Action:**

The system must allow an Admin to remove user-submitted content (ratings and reviews) that have been flagged as inappropriate. This function is designed specifically for Admin roles and is not accessible to Regular Users.
- **Requirements/Inputs:**
 - Admin's secure login credentials to access the moderation panel.
 - A list of flagged content items along with details (user submission).
 - Input selection for the admin to delete the user content.
- **Source:**
 - Administrative interface and automated flagging mechanisms.
- **Pre-condition:**
 - The user is authenticated as an Admin with the necessary privileges.
 - There exists one or more submissions by users.
- **Post-condition:**
 - The selected content is removed from display.

- The system logs the admin's action for auditing purposes.
 - **Output:**
 - A confirmation message for each moderation action (e.g., "Content removed") and an updated view of public content reflecting the changes.
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4. Non-Functional Requirements

1. Performance

- The application should launch within **5 seconds** on standard hardware
- The system shall respond to search queries within 10 seconds.

2. Security

- All passwords shall be stored using **secure hashing**.
- Only admins can access the admin panel, enforced via **role-based access control**.

3. Usability

- Response time for UI actions should be **under 200ms** for a smooth experience
- The system shall provide clear, concise error messages with actionable instructions when issues occur.

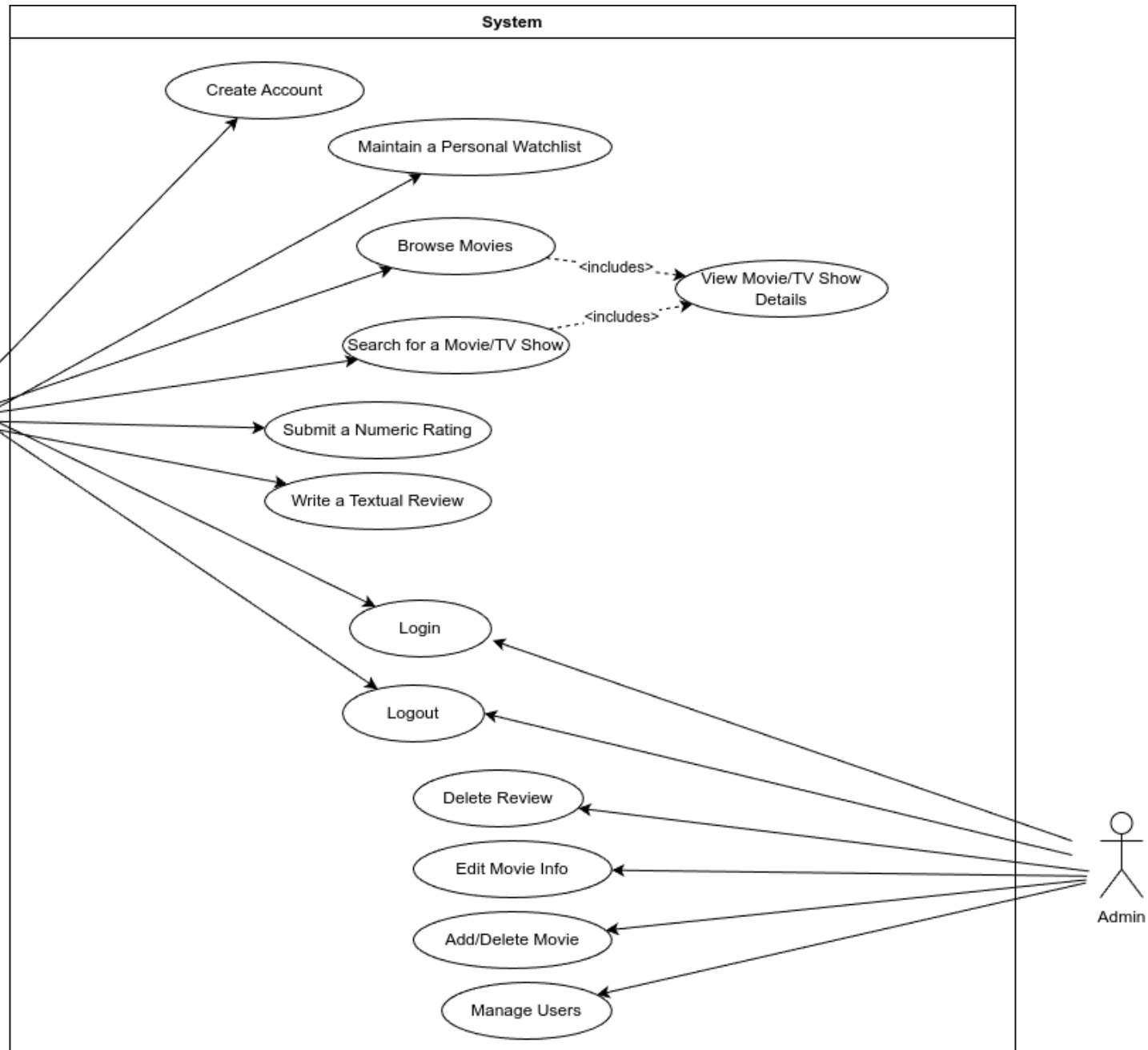
4. Reliability

- The system shall be able to handle at least 5 concurrent users.

5. Maintainability

- The system should follow **Object-Oriented Programming (OOP) and SOLID principles** to result in clean, reusable code.
- The codebase shall follow **standard design patterns** to facilitate updates and debugging.
- The system shall be modular, enabling developers to add new features without impacting existing functionality.

Use Case Diagram



Sequence Diagram

For the "Submit a Numeric Rating" Use case.

